



Date :	05.06.25	Horaire :	08:15 - 11:00	Durée :	165 minutes
Discipline :	MATHE - ANALY	Type :	écrit	Section(s) :	CC / CC-4LANG / CI
Numéro du candidat :					

Question 1:

(1+3)+(4+5)=13 points

1. a) $\forall a \in \mathbb{R}_+^* \setminus \{1\}; \forall x, y \in \mathbb{R}_+^*$ $\log_a \left(\frac{x}{y} \right) = \log_a x - \log_a y.$

voir page 55 EM66.

b) $\forall a \in \mathbb{R}_+^* \setminus \{1\}; \forall x \in \mathbb{R}_+^*; \forall r \in \mathbb{R}$ $\log_a (x^r) = r \cdot \log_a x$

voir page 55 EM66.

2. a) $2^{x+1} + 7 = 2^{2-x}$ $C.E : x \in \mathbb{R}$

$$\iff 2 \cdot 2^x + 7 = \frac{2^2}{2^x}$$

$$\iff 2 \cdot 2^{2x} + 7 \cdot 2^x = 4$$

$$\iff 2 \cdot 2^{2x} + 7 \cdot 2^x - 4 = 0$$

$$\iff \begin{cases} y = 2^x \\ 2y^2 + 7y - 4 = 0 \end{cases}$$

$$\Delta = 81; \sqrt{\Delta} = 9; \alpha = -4; \beta = \frac{1}{2}$$

$$\iff \underbrace{2^x}_{>0} = \underbrace{-4}_{<0} \text{ impossible ou } 2^x = 2^{-1}$$

$$\iff x = -1$$

donc $S = \{-1\}$

b) $\log_2 (x-2) + \log_{\frac{1}{2}} (x-1) \leq 2 - \frac{1}{2} \cdot \log_{\sqrt{2}} (x+5)$

$$C.E : \begin{cases} x-2 > 0 \\ x-1 > 0 \\ x+5 > 0 \end{cases} \iff \begin{cases} x > 2 \\ x > 1 \\ x > -5 \end{cases} \text{ donc } x \in]2; +\infty[$$

$$\log_2 (x-2) + \log_{\frac{1}{2}} (x-1) \leq 2 - \frac{1}{2} \cdot \log_{\sqrt{2}} (x+5)$$

$$\iff \log_2 (x-2) + \frac{\log_2 (x-1)}{\log_2 (\frac{1}{2})} \leq 2 - \frac{1}{2} \cdot \frac{\log_2 (x+5)}{\log_2 (\sqrt{2})}$$

$$\iff \log_2 (x-2) - \log_2 (x-1) \leq 2 - \log_2 (x+5)$$

$$\iff \log_2 (x-2) + \log_2 (x+5) \leq \log_2 (2^2) + \log_2 (x-1)$$

$$\iff \log_2 ((x-2) \cdot (x+5)) \leq \log_2 (4 \cdot (x-1)) \quad \log_2 \text{ est une bijection } \nearrow$$

$$\iff x^2 + 3x - 10 \leq 4x - 4$$

$$\iff x^2 - x - 6 \leq 0 \quad \Delta = 25; \sqrt{\Delta} = 5; \alpha = -2; \beta = 3; (\quad + \quad (-2) \quad - \quad 3 \quad + \quad)$$

$$\iff x \in [-2; 3]$$

donc $S =]2; 3]$

Question 2:

3+1+3+2+4=13 points

Soit la fonction f définie par:

$$f : x \mapsto -e^{-x} (x^2 + 4x + 5)$$

1. $\text{dom}(f) = \mathbb{R}$

$$\lim_{x \rightarrow +\infty} \underbrace{-e^{-x}}_{\rightarrow 0} \cdot \underbrace{(x^2 + 4x + 5)}_{\rightarrow +\infty} = \lim_{x \rightarrow +\infty} - \frac{\overbrace{x^2 + 4x + 5}^{\rightarrow +\infty}}{\underbrace{e^x}_{\rightarrow +\infty}} \stackrel{2 \times H}{=} \lim_{x \rightarrow +\infty} \frac{-2}{\underbrace{e^x}_{\rightarrow +\infty}} = 0$$

donc $A.H \equiv y = 0$ en $-\infty$.

$$\lim_{x \rightarrow -\infty} \underbrace{-e^{-x}}_{\rightarrow -\infty} \cdot \underbrace{(x^2 + 4x + 5)}_{\rightarrow +\infty} = -\infty \text{ donc } A.O \text{ possible en } -\infty$$

$$\lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \lim_{x \rightarrow -\infty} \frac{-e^{-x} (x^2 + 4x + 5)}{x} = \lim_{x \rightarrow -\infty} \underbrace{-e^{-x}}_{\rightarrow -\infty} \cdot \underbrace{\left(x + 4 + \frac{5}{x}\right)}_{\rightarrow -\infty} = +\infty$$

donc pas d' $A.O$ en $+\infty$ ($B.P$ de direction asymptotique (Oy) en $+\infty$)

2. Position de $\mathcal{C}_f$ par rapport à l' $A.H$.

$$\begin{aligned} f(x) - 0 \geq 0 &\iff -e^{-x} (x^2 + 4x + 5) \geq 0 & | \cdot (-e^x) \\ &\iff x^2 + 4x + 5 \leq 0 & \Delta = -4 < 0 \\ &\iff \underbrace{x^2 + 4x + 5}_{>0} \leq 0 & \text{impossible} \end{aligned}$$

donc $\forall x \in \mathbb{R}$ $\mathcal{C}_f$ est en-dessous de l' $A.H$.

3. $\text{dom}(f') = \mathbb{R}$

$$\begin{aligned} f'(x) &= (-e^{-x} (x^2 + 4x + 5))' \\ &= e^{-x} (x^2 + 4x + 5) - e^{-x} (2x + 4) \\ &= e^{-x} (x^2 + 2x + 1) \\ &= (x + 1)^2 \cdot e^{-x} \end{aligned}$$

$$f'(x) = 0 \iff (x + 1)^2 = 0 \iff x = -1$$

tableau de variation

x	$-\infty$		-1		$+\infty$
$f'(x)$		$+$	0		$+$
$f(x)$		$\nearrow$	$f(-1) = -2e$	$\nearrow$	

4. $t_1 \equiv y = f'(1) \cdot (x - 1) + f(1)$

On a: $f(1) = -10e^{-1} = -\frac{10}{e}$; $f'(1) = 4e^{-1} = \frac{4}{e}$

ainsi $t_1 \equiv y = \frac{4}{e} \cdot (x - 1) - \frac{10}{e} \iff y = \frac{4}{e} \cdot x - \frac{14}{e}$

5. $\text{dom}(f'') = \mathbb{R}$

$$\begin{aligned} f''(x) &= ((x+1)^2 \cdot e^{-x})' \\ &= 2(x+1) \cdot e^{-x} - (x+1)^2 \cdot e^{-x} \\ &= e^{-x} \cdot (2x+2 - x^2 - 2x - 1) \\ &= (1-x^2) \cdot e^{-x} \end{aligned}$$

$$f''(x) = 0 \iff 1 - x^2 = 0 \iff x = -1 \text{ ou } x = 1$$

tableau de concavité

x	$-\infty$		-1		1		$+\infty$
$f''(x)$		$-$	0	$+$	0	$-$	
$\mathcal{C}_f$		concave	P.I	convexe	P.I	concave	

$$A(-1; f(-1)) = A(-1; -2e) \text{ et } B(1; f(1)) = B(1; -\frac{10}{e})$$

sont les points d'inflexion de $\mathcal{C}_f$.

Question 3:

4+2=6 points

1. $\lim_{x \rightarrow +\infty} \left(\frac{x-3}{x-1} \right)^{2x+1} = \lim_{x \rightarrow +\infty} \left(1 - \frac{2}{x-1} \right)^{2x+1}$ "1 ∞ " f.i

On pose: $\frac{-2}{x-1} = \frac{1}{y} \iff -2y = x-1 \iff x = 1-2y$ on a: $x \rightarrow +\infty \iff y \rightarrow -\infty$

$$\begin{aligned} \lim_{x \rightarrow +\infty} \left(1 - \frac{2}{x-1} \right)^{2x+1} &= \lim_{y \rightarrow -\infty} \left(1 + \frac{1}{y} \right)^{3-4y} \\ &= \left(\lim_{y \rightarrow -\infty} \left(1 + \frac{1}{y} \right)^y \right)^{-4} \cdot \left(\lim_{y \rightarrow -\infty} \left(1 + \frac{1}{y} \right) \right)^3 \\ &= e^{-4} \end{aligned}$$

$$\begin{aligned}
2. \quad \lim_{x \rightarrow +\infty} \left(x^2 \cdot \operatorname{Arcsin} \left(\frac{1}{x} \right) \right) &= \lim_{x \rightarrow +\infty} \left(\underbrace{x^2}_{\rightarrow +\infty} \cdot \underbrace{\operatorname{Arcsin} \left(\frac{1}{x} \right)}_{\rightarrow 0} \right) \quad f.i. \\
\lim_{x \rightarrow +\infty} \left(x^2 \cdot \operatorname{Arcsin} \left(\frac{1}{x} \right) \right) &= \lim_{x \rightarrow +\infty} \frac{\overbrace{\operatorname{Arcsin} \left(\frac{1}{x} \right)}^{\rightarrow 0}}{\underbrace{\frac{1}{x^2}}_{\rightarrow 0}} \\
&= \lim_{x \rightarrow +\infty} \frac{\frac{-\frac{1}{x^2}}{\sqrt{1 - \frac{1}{x^2}}}}{\frac{-2}{x^3}} \\
&= \lim_{x \rightarrow +\infty} \frac{\overbrace{x}^{\rightarrow +\infty}}{\underbrace{2\sqrt{\frac{x^2 - 1}{x^2}}}_{\rightarrow 2}} \\
&= +\infty
\end{aligned}$$

Question 4:

8 points

Soit f la fonction définie par:

$$f : x \mapsto -3x + \ln \left(\frac{2x}{x-2} \right)$$

$$C.E : \frac{2x}{x-2} > 0$$

x	$-\infty$		0		2		$+\infty$
N		-	0	+		+	
D		-		-	0	+	
Q		+	0	-		+	

donc $\operatorname{dom}(f) =]-\infty; 0[\cup]2; +\infty[$

$$\lim_{x \rightarrow +\infty} \left(\underbrace{-3x}_{\rightarrow -\infty} + \underbrace{\ln \left(\frac{2x}{x-2} \right)}_{\rightarrow \ln 2} \right) = -\infty \quad A.O \text{ possible}$$

$$\text{On a: } \lim_{x \rightarrow +\infty} \frac{2x}{x-2} = \lim_{x \rightarrow +\infty} \frac{2x}{x} = 2$$

 $y = -3x + \ln 2$ est A.O à C_f en $+\infty$ car

$$f(x) - (-3x + \ln 2) = \ln \left(\frac{2x}{x-2} \right) - \ln 2 = \ln \left(\frac{2x}{2(x-2)} \right) = \ln \left(\frac{x}{x-2} \right)$$

$$\text{et } \lim_{x \rightarrow +\infty} \ln \left(\underbrace{\frac{x}{x-2}}_{\rightarrow 1} \right) = 0$$

Position $\mathcal{C}_f$ par rapport à $A.O$.

$$f(x) - (-3x + \ln 2) = \ln \left(\frac{x}{x-2} \right)$$

$$\ln \left(\frac{x}{x-2} \right) > 0 \iff \frac{x}{x-2} > 1 \iff \frac{2}{x-2} > 0$$

Si $x \in \mathbb{R}_-^*$ alors $\mathcal{C}_f$ est en-dessous de l' $A.O$ et si $x > 2$ alors $\mathcal{C}_f$ est au-dessus de l' $A.O$.

Question 5:

4+(3+4)=11 points

1. Calculons d'abord $J = \int 15x\sqrt{2-x} \, dx$

$$I.P.P : a(x) = 15x; b'(x) = \sqrt{2-x}; a'(x) = 15; b(x) = -\frac{2}{3}(2-x)^{\frac{3}{2}}$$

$$\begin{aligned} \text{donc } J &= -10x \cdot (2-x)^{\frac{3}{2}} + 10 \cdot \int (2-x)^{\frac{3}{2}} \, dx \\ &= -10x \cdot (2-x)^{\frac{3}{2}} - 10 \cdot \frac{2}{5} (2-x)^{\frac{5}{2}} + k \quad k \in \mathbb{R} \\ &= -10x(2-x)^{\frac{3}{2}} - 4(2-x)^{\frac{5}{2}} + k \end{aligned}$$

$$\begin{aligned} \text{Ainsi } I &= \left[-10x(2-x)^{\frac{3}{2}} \right]_1^{-2} - \left[4(2-x)^{\frac{5}{2}} \right]_1^{-2} \\ &= (20 \cdot 8 - (-10)) - (4 \cdot 32 - 4) \\ &= 46 \end{aligned}$$

$$2. \quad a) \quad \frac{10x^2 - 8x + 4}{(1 - 2x)(1 + x^2)} = \frac{a}{1 - 2x} + \frac{bx + c}{1 + x^2} \quad \forall x \in \mathbb{R} \setminus \{\frac{1}{2}\}$$

$$\iff 10x^2 - 8x + 4 = a \cdot (1 + x^2) + (bx + c) \cdot (1 - 2x)$$

$$\iff 10x^2 - 8x + 4 = a + ax^2 + bx - 2bx^2 + c - 2cx$$

$$\iff 10x^2 - 8x + 4 = (a - 2b)x^2 + (b - 2c)x + a + c$$

$$\iff \begin{cases} a - 2b = 10 \\ b - 2c = -8 \\ a + c = 4 \end{cases}$$

$$\iff \begin{cases} a = 2b + 10 \\ b - 2c = -8 \\ 2b + 10 + c = 4 \end{cases}$$

$$\iff \begin{cases} a = 2b + 10 \\ b - 2c = -8 & (2) \\ 2b + c = -6 & (3) \end{cases}$$

$$\iff \begin{cases} a = 2b + 10 \\ b - 2c = -8 & (2) \\ 5b = -20 & 2 \cdot (3) + (2) \end{cases}$$

$$\iff \begin{cases} a = 2 \\ c = 2 \\ b = -4 \end{cases}$$

$$\text{donc } f(x) = \frac{2}{1 - 2x} + \frac{-4x + 2}{1 + x^2} \quad \forall x \in \mathbb{R} \setminus \{\frac{1}{2}\}$$

$$\begin{aligned} b) \quad \int f(x) \, dx &= \int \left(\frac{2}{1 - 2x} + \frac{-4x + 2}{1 + x^2} \right) dx \\ &= \int \frac{2}{1 - 2x} dx - \int \frac{4x}{1 + x^2} dx + \int \frac{2}{1 + x^2} dx \\ &= -\ln|1 - 2x| - 2\ln(1 + x^2) + 2\arctan x + k \end{aligned}$$

$$\text{donc } F(x) = -\ln|1 - 2x| - 2\ln(1 + x^2) + 2\arctan x + k$$

$$F(1) = 0 \iff -\ln|-1| - 2\ln(2) + 2\arctan(1) + k = 0$$

$$\iff -2\ln 2 + \frac{1}{2}\pi + k = 0$$

$$\iff k = 2\ln 2 - \frac{\pi}{2}$$

$$\text{donc } F(x) = -\ln|1 - 2x| - 2\ln(1 + x^2) + 2\arctan x + 2\ln 2 - \frac{\pi}{2} \quad \forall x \in]\frac{1}{2}; +\infty[$$

Question 6:

(3+1+2)+3=9 points

$$\begin{aligned}
 1. \quad a) \quad f(x) \geq g(x) &\iff \frac{(2 + \ln(x))^2}{x} \geq \frac{4 + 2 \ln(x)}{x} \\
 &\iff \frac{4 + 4 \ln(x) + \ln^2(x) - 4 - 2 \ln(x)}{x} \geq 0 \\
 &\iff \frac{\ln(x)(\ln(x) + 2)}{x} \geq 0
 \end{aligned}$$

x	0		e^{-2}		1		$+\infty$
$\ln(x)$		-		-	0	+	
$\ln(x) + 2$		-	0	+		+	
$f(x) - g(x)$		+	0	-	0	+	

- b) $\forall x \in [e^{-2}; 1]$ on a: $f(x) \geq g(x)$
 $\forall x \in]0; e^{-2}] \cup [1; +\infty[$ on a: $g(x) \geq f(x)$

$$\begin{aligned}
 c) \quad \mathcal{A} &= \int_{e^{-2}}^1 (g(x) - f(x)) \, dx + \int_1^e (f(x) - g(x)) \, dx \\
 &= \int_{e^{-2}}^1 \frac{-2 \ln(x) - \ln^2(x)}{x} \, dx + \int_1^e \frac{2 \ln(x) + \ln^2(x)}{x} \, dx \\
 &= \int_{e^{-2}}^1 \left(-2 \cdot \frac{\ln(x)}{x} - \frac{\ln^2(x)}{x} \right) \, dx + \int_1^e \left(2 \cdot \frac{\ln(x)}{x} + \frac{\ln^2(x)}{x} \right) \, dx \\
 &= \left[-\ln^2(x) - \frac{\ln^3(x)}{3} \right]_{e^{-2}}^1 + \left[\ln^2(x) + \frac{\ln^3(x)}{3} \right]_1^e \\
 &= \frac{4}{3} + \frac{4}{3} \\
 &= \frac{8}{3} \text{ u.a}
 \end{aligned}$$

2. Sur le graphique on voit que $f(x) = g(x) \iff x = -1$ ou $x = 2$.

$$\forall x \in [-1; 2] \quad f(x) \leq g(x) \leq 0$$

$$\begin{aligned}
 \mathcal{V} &= \pi \cdot \int_{-1}^2 f^2(x) \, dx - \pi \int_{-1}^2 g^2(x) \, dx \\
 &= \pi \cdot \int_{-1}^2 (x^2 - 5)^2 \, dx - \pi \int_{-1}^2 (x - 3)^2 \, dx \\
 &= \pi \cdot \left[\frac{1}{5}x^5 - \frac{10}{3}x^3 + 25x \right]_{-1}^2 - \pi \cdot \left[\frac{1}{3}(x - 3)^3 \right]_{-1}^2 \\
 &= \pi \cdot \frac{258}{5} - \pi \cdot 21 \\
 &= \frac{153}{5} \pi \text{ u.v.}
 \end{aligned}$$